Case study : "The Attack on BrightMask Online". 27/08/2026

Topic covered

- Overview of cybersecurity
- Threats and Vulnerabilities
- Security principles and concepts
- Common Attacks and Their impact

Learning objective

By the end of the activity, students should be able to :

1. Identify cybersecurity threats in a real-world situation
2. Distinguish between a threat and a vulnerability
3. Apply the CIA Triad - Confidentiality, Integrity and Availability.
4. Identify common cyberattacks.
5. Analyze the impact of an attack on an organization.
6. Suggest basic security measures.

Case Study Scenario - BrightMart is a medium-sized online shopping company. customers use its website and mobile application to purchase products.

The Company stores :

- customers names
- email addresses
- phone numbers
- Delivery addresses
- passwords
- order information
- Payment-related information

BrightMart has around 100 employees.

The company recently hired several new employees. To make things easier, the IT department gave employees access to different internal systems.

However, the company has the following security weaknesses :

- some employeess use simple passwords such as BrightMart123
- Many employees use the same passwords for multiple systems.
- Software updates are sometimes delayed.
- The company does not regularly conduct employee cybersecurity training.
- The company does not use multi-factor authenication for some important accounts.
- Employees can access sensitive information that they do not necessarily need for

their jobs.

- Their Backups are performed only once a month.

The Incident

One monday morning, an employee named Rahul receives an email.

Subject : Urgent — Your account will be Suspended.

The email appears to come from the company's IT department.

It says :-

"Your company account will be suspended today. Click the link below and verify your password immediately"

Rahul clicks the link.

The page looks almost identical to the company's login page.

He enters his username and password.

Rahul does not realize that the website is controlled by an attacker.

The attacker now has Rahul's login credentials.

Stage 1 - Unauthorized Access

The attacker uses Rahul's credentials to access BrightMark's internal system.

Because Rahul uses the same password for another system, the attacker is also able to access that system.

The attacker discovers that Rahul has access to customer information

The attacker downloads a large amount of customer data.

Q. What cybersecurity problem has occured?

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Answers • Phishing • Credential theft • Unauthorized access
• Data breach

Stage 2 - Malware Attack

The attacker sends another email to Rahul containing a malicious attachment.
Rahul open the attachment.
Malware is installed on his computer.
The Malware begins communicating with the attacker's system.
The attacker now attempts to move from Rahul's computer to other system within BrightMart's Network.

Q. Which Vulnerability allowed the attacker to move further into the organization?

Answer Students should identify possibilities such as:
• weak passwords • Excessive user privileges • poor network security
• Lack of MFA • Lack of employee awareness • Outdated software

Stage 3 - Ransomware

Two days later, employees arrive at work and discover that they cannot access important files.
A messages appears on their computers:
"Your files have been encrypted. Pay the ransom to recover them".
The company's customer database and several important business files are encrypted.
Unfortunately, the company's most recent backup is almost a month old.
BrightMart's Website stops working.
Customers cannot place orders.
The customers support team cannot access customer records.

Stage 4 - Business Impact

BrightMart now faces several problems.
Financial Impact - The company loses sales because its website is unavailable.
Data Impact - Customer information may have been stolen.

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Operational Impact
Reputation Impact
Legal | Compliance I
consequences de
which laws apply

Group 1 - Identif
Q. What threats
Answers They r
• phishing • Ma
• Unauthorized
Group 2 - Identif
Q. What weakness
Answers Vulne
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No MFA
Lack of training
Delayed updat
Excessive Privile
Infrequent backup

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Operational Impact - Employees cannot access important systems.
Reputation Impact - Customers lose confidence in the company.
Legal / Compliance Impact - The company may face regulatory or legal consequences depending on what information was exposed and which laws apply.

Group 1 - Identify The Threats
Q. What threats are present in this case?
Answers They may identify:
· phishing · Malware · Ransomware · Credential theft
· Unauthorized access · Data theft

Group 2 - Identify the Vulnerabilities
Q. What weakness in BrightMart's security allowed the attack to succeed?

Answers	Vulnerability	How it helped the attacker
	Weak passwords	Easier to compromise accounts
	Password reuse	One stolen password may be enough to given access to multiple systems.
	No MFA	Stolen password may be enough to log in.
	Lack of training	Employee falls for phishing.
	Delayed updates	Vulnerable software may be exploited.
	Excessive Privileges	Attacker can access more data.
	Infrequent backups	Difficult to recover after ransomware.

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Group 3 - Apply the CIA Triad?

Q1. Was Confidential information exposed?

Answers Yes. Customer information was accessed / downloaded by the attacker.

Q2. Was Information changed or damaged?

Answers potentially yes. The attacker could modify or encrypt files.

Q3. Were systems available to legitimate users?

Answers No. The website and internal systems became unavailable because of ransomware.

so, Confidentially → Affected

Integrity → Affected

Availability → Affected

Group 4 - Security Solutions

Q1. Who is responsible for cybersecurity?

Answers Expected discussion: Not only the IT department. Every employee who uses the organization's system has a role in cybersecurity.

Q2. Was phishing the only problem?

Answers No. Phishing was the inital attack, but weakness such as Password reuse, lack of MFA, excessive privileges, poor backups, and insufficient training allowed the attack to become much more damaging.

Q3. What was the biggest Vulnerability?

Answers There is no single correct answer.

Students should justify their choice.

For example:- "Lack of employee awareness was a major vulnerability because the attack started when Rahul clicked the phising link".

Another student might argue:

"Poor Backup practices were the biggest problem because BrightMart could not quickly recover from ransomware".

Both can be valid if properly justified.

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Quick Assessment

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Q3. Which three se

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Q4. What was the i

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Q5. Suggest three

Ans 1. Multi factor
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Quick Assessment

Q1. What was the initial attack against BrightMart?

Ans The initial attack was a phishing attack. Rahul received a fake email that looked like it came from IT department.

Q2. Give three vulnerabilities that helped the attacker?

Ans 1. Password Reuse 2. Weak passwords 3. Delayed Updates.

Q3. Which three security principles of the CIA Triad were affected?

Ans Confidentiality, Integrity, Availability

Q4. What was the impact of ransomware on BrightMart?

Ans The ransome encrypted the customer database and important business files. This caused financial losses, opertional problems, data lose

Q5. Suggest three security controls BrightMart should implement?

ANS 1. Multi factor authentication (MFA) for important accounts.

2. Regular cybersecurity and phishing awareness training for employees.

3. Frequent and seaure backups of important data, rather than only once a month.

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